Homework - 4

(1) $D^{(k)}[i][j]$ = length of shortest paths from vertex v_i to v_j using only vertices from the set $\{v_1, v_2, v_3, \ldots, v_k\}$.

Initially, $D^{(0)}$ is the weighted adjacency matrix W.

$$D^{(0)} = \begin{bmatrix} 0 & 4 & \infty & \infty & \infty & 10 & \infty \\ 3 & 0 & \infty & 18 & \infty & \infty & \infty \\ \infty & 6 & 0 & \infty & \infty & \infty & \infty \\ \infty & 5 & 15 & 0 & 2 & 19 & 5 \\ \infty & \infty & 12 & 1 & 0 & \infty & \infty \\ \infty & \infty & \infty & 8 & \infty & \infty & 0 \end{bmatrix}$$

* iteration -1 :- Paths through v_1

$D[i][j] = \min(D[i][j],\ D[i][1] + D[1][j])$

$\rightarrow$ $D[2][6] = \min(\infty, 3+10) = 13$

$$D^{(1)} = \begin{bmatrix} 0 & 4 & \infty & \infty & \infty & 10 & \infty \\ 3 & 0 & \infty & 18 & \infty & 13 & \infty \\ \infty & 6 & 0 & \infty & \infty & \infty & \infty \\ \infty & 5 & 15 & 0 & 2 & 19 & 5 \\ \infty & \infty & 12 & 1 & 0 & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty & 0 & 10 \\ \infty & \infty & \infty & 8 & \infty & \infty & 0 \end{bmatrix}$$

* <u>iteration-2</u> :- paths that have v_2

→ $D[1][4] = \min(D[1][4], D[1][2] + D[2][4])$

$= \min(\infty, 4+18) = 22$

→ $D[3][1] = \min(\infty, 6+3) = 9$

→ $D[3][4] = \min(\infty, 6+18) = 24$

→ $D[3][6] = \min(\infty, 6+13) = 19$

→ $D[4][1] = \min(\infty, 5+3) = 8$

→ $D[4][6] = \min(19, 5+13) = 18$

$$D^{(2)} = \begin{bmatrix} 0 & 4 & \infty & 22 & \infty & 10 & \infty \\ 3 & 0 & \infty & 18 & \infty & 13 & \infty \\ 9 & 6 & 0 & 24 & \infty & 19 & \infty \\ 8 & 5 & 15 & 0 & 2 & 18 & 5 \\ \infty & \infty & 12 & 1 & 0 & \infty & \infty \\ \infty & \infty & \infty & \infty & \infty & 0 & 10 \\ \infty & \infty & \infty & 8 & \infty & \infty & 0 \end{bmatrix}$$

* <u>iteration-3</u> :- paths through v_3

→ $D[5][1] = \min(D[5][1], D[5][3] + D[3][1])$

$= \min(\infty, 12+9) = 21$

→ $D[5][2] = \min(\infty, 12+6) = 18$

→ $D[5][6] = \min(\infty, 12+19) = 31$

$$D^{(3)} = \begin{bmatrix} 0 & 4 & \infty & 22 & \infty & 10 & \infty \\ 3 & 0 & \infty & 18 & \infty & 13 & \infty \\ 9 & 6 & 0 & 24 & \infty & 19 & \infty \\ 8 & 5 & 15 & 0 & 2 & 18 & 5 \\ 21 & 18 & 12 & 1 & 0 & 31 & \infty \\ \infty & \infty & \infty & \infty & \infty & 0 & 10 \\ \infty & \infty & \infty & 8 & \infty & \infty & 0 \end{bmatrix}$$

* <u>iteration-4</u> :- paths through v_4

→ D[1][3] = 37

D[1][5] = 24

D[1][7] = 27

→ D[2][3] = 33

D[2][5] = 20

D[2][7] = 23

→ D[3][5] = 26

D[3][7] = 29

→ D[5][1] = 9

D[5][2] = 6

D[5][6] = 19

D[5][7] = 6

→ D[7][1] = 16

D[7][2] = 13

D[7][3] = 23

D[7][5] = 10

D[7][6] = 26

$$D^{(4)} = \begin{bmatrix} 0 & 4 & 37 & 22 & 24 & 10 & 27 \\ 3 & 0 & 33 & 18 & 20 & 13 & 23 \\ 9 & 6 & 0 & 24 & 26 & 19 & 29 \\ 8 & 5 & 15 & 0 & 2 & 18 & 5 \\ 9 & 6 & 12 & 1 & 0 & 19 & 6 \\ \infty & \infty & \infty & \infty & \infty & 0 & 10 \\ 16 & 13 & 23 & 8 & 10 & 26 & 0 \end{bmatrix}$$

* iteration - 5 : paths through v_5

→ $D[1][3] = \min(37, 24+12) = 36$

→ $D[2][3] = \min(33, 20+12) = 32$

→ $D[4][3] = \min(15, 2+12) = 14$

→ $D[7][3] = \min(23, 10+12) = 22$

$$D^{(5)} = \begin{bmatrix} 0 & 4 & 36 & 22 & 24 & 10 & 27 \\ 3 & 0 & 32 & 18 & 20 & 13 & 23 \\ 9 & 6 & 0 & 24 & 26 & 19 & 29 \\ 8 & 5 & 14 & 0 & 2 & 18 & 5 \\ 9 & 6 & 12 & 1 & 0 & 19 & 6 \\ \infty & \infty & \infty & \infty & \infty & 0 & 10 \\ 16 & 13 & 22 & 8 & 10 & 26 & 0 \end{bmatrix}$$

* iteration-6 :- paths through v_6

$D[1][7] = \min(27, 10+10) = 20$

$$D^{(6)} = \begin{bmatrix} 0 & 4 & 36 & 22 & 24 & 10 & 20 \\ 3 & 0 & 32 & 18 & 20 & 13 & 23 \\ 9 & 6 & 0 & 24 & 26 & 19 & 29 \\ 8 & 5 & 14 & 0 & 2 & 18 & 5 \\ 9 & 6 & 12 & 1 & 0 & 19 & 6 \\ \infty & \infty & \infty & \infty & \infty & 0 & 10 \\ 16 & 13 & 22 & 8 & 10 & 26 & 0 \end{bmatrix}$$

* iteration-7 :- paths through v_7

→ $D[6][1] = \min(\infty, 10+16) = 26$

→ $D[6][2] = \min(\infty, 10+13) = 23$

→ $D[6][3] = \min(\infty, 10+22) = 32$

→ $D[6][4] = \min(\infty, 10+8) = 18$

→ $D[6][5] = \min(\infty, 10+10) = 20$

$$D^{(7)} = \begin{bmatrix} 0 & 4 & 36 & 22 & 24 & 10 & 20 \\ 3 & 0 & 32 & 18 & 20 & 13 & 23 \\ 9 & 6 & 0 & 24 & 26 & 19 & 29 \\ 8 & 5 & 14 & 0 & 2 & 18 & 5 \\ 9 & 6 & 12 & 1 & 0 & 19 & 6 \\ 26 & 23 & 32 & 18 & 20 & 0 & 10 \\ 16 & 13 & 22 & 8 & 10 & 26 & 0 \end{bmatrix}$$

↓

final distance matrix

2. print shortest path algorithm from V_7 to V_3.

→ $P[7][3] = 5$

$P[7][5] = 4$

$P[7][4] = 0$

$P[4][5] = 0$

$P[5][3] = 0$

These are the P values for the route.

→ Step-by-step actions :-

(i) Call path(7,3)

$P[7][3] = 5 \neq 0$ then,

1. call Path(7,5)
2. print V_5
3. call path(7,3)

(ii) Expand path(7,5)

$P[7][5] = 4 \neq 0$ then,

1. call path(7,4)
2. Print V_4
3. call path(4,5)

(iii) expand path(7,4)

- $P[7][4] = 0$ → we don't execute next lines
- V_4 gets printed → from step-2 in (ii)
- expand Path(4,5) → from step-3 in (ii)
  $P[4][5] = 0$ → we don't execute next lines
- Now we go to steps-2,3 in (i)
- V_5 gets printed → from step-2 in (i)

- expand path(5,3) : → from step-3 in (i)

  $P[5][3] = 0$, we don't execute next lines.

Final Output Printed:

V_4

V_5

$\therefore$ Shortest path from V_7 to V_3 is $V_7 \rightarrow V_4 \rightarrow V_5 \rightarrow V_3$

(3) Different binary search trees using six distinct keys:

→ If we sort these n distinct keys and select the i^{th} key as the root of the tree, then we must have $i-1$ smaller keys to the left subtree and $n-i$ larger keys to the right subtree.

→ The total number of unique trees is the sum of possible combinations for each root i

$$C_n = \sum_{i=1}^{n} (C_{i-1} \times C_{n-i})$$

here C_n is the n^{th} Catalan number

$$C_n = \sum_{i=1}^{n} (C_{i-1} \times C_{n-i})$$

Another formula for C_n is

$$C_n = \frac{1}{(n+1)} \, {}^{2n}C_n = \frac{1}{(n+1)} \binom{2n}{n}$$

$$C_n = \frac{1}{n+1} \cdot \frac{(2n)!}{n! \times n!}$$

In this case $n=6$

$$C_6 = \frac{1}{\cancel{7}} \times \frac{\cancel{12} \times 11 \times \cancel{10} \times \overset{3}{\cancel{9}} \times \overset{4}{\cancel{8}} \times \cancel{7}}{\cancel{18} \times \cancel{120}} = \underline{\underline{132}}$$ unique binary search trees.

(4) CASE - 0.05

ELSE - 0.15

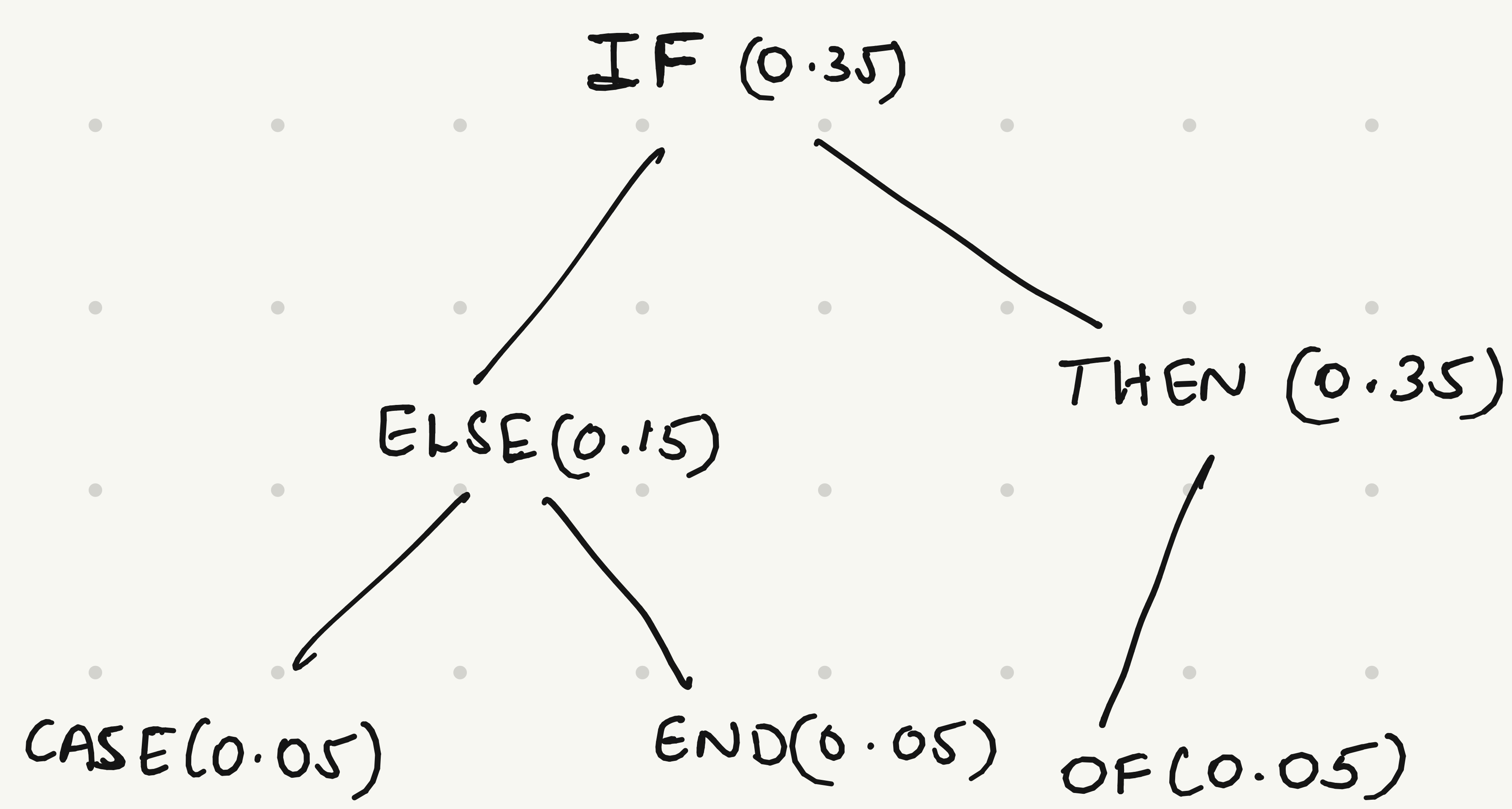
END - 0.05

IF - 0.35

OF - 0.05

THEN - 0.35

→ OBST should have the minimal expected search cost.
High probability values at the top, low probability values at the bottom.



expected search cost $= 1 \times 0.35 + 2(0.15 + 0.35) + 3(0.05 + 0.05 + 0.05)$

$= 0.35 + 1 + 0.45$

$= \underline{\underline{1.8}}$